

# Eros Gulo

P. Eng., M. Sc., B. Eng.

(416) 414 - 8044

erosgulo@gmail.com

linkedin.com/in/erosgulo

**Senior Software Engineer**

**Research & Development**

Vaughan

(Greater Toronto Area)

Ontario, Canada

## Professional Summary

Professional Engineer with 9 years of experience in R&D and software engineering in large enterprise and start-up companies. I have a proven track record of developing server and client software projects in .NET C#, C++, Java, Kotlin, and Swift from ideation through to their timely completion.

As an R&D Engineer at Mapsted, I spearheaded several new features and technologies in the realm of mapping, positioning and navigation on mobile devices (Android and iOS). I collaborated with a cross-functional team on design, client and server software engineering, UX, testing and troubleshooting. I also oversaw deployments around the world, coordinating coworkers, partners and clients in the deployment process, as well as providing technical support and troubleshooting both remotely and on-site, as needed.

As an Innovation Engineer at Fugro, I developed a multi-sensor surveying system (comprised of cameras, LiDAR, INS and other sensors) operating in a vehicle environment. Alongside the development of the main system with my team, I also developed software for camera calibration, processing and geo-referencing data from sensors, asset positioning using imagery (photogrammetry) and LiDAR point colorization.

Having worked in multiple cross-functional and geographically distributed teams, I am versatile, proactive and excited for a new challenge.

## Technical Skills and Competencies

- Professional Engineer (P. Eng.) licensed by Professional Engineers Ontario
- Sensor Calibration (Cameras, LiDAR, INS, IMU, GNSS)
- Sensor Data Georeferencing / Geolocation
- Location Services (Mapping, Positioning, Routing, Navigation)
- Algorithm Design and Development
- Object Oriented Design and Programming (OOD / OOP)
- Languages: C# (Proficient), C++ (Proficient), Java, Kotlin, Swift, Python, SQL
- AI Coding Assistants (GitHub Copilot, Cursor)
- Agile Scrum Software Development
- Continuous Integration and Continuous Delivery (CI / CD)
- Git Version / Source Control (Proficient)
- Microsoft Visual Studio (Proficient), .NET 6 / 8 SDK (Proficient)
- Android Studio (Proficient), Android SDK and NDK (Proficient)
- Apple Xcode, iOS SDK (Core Location and Core Motion)
- NuGet Package Manager (Using and Publishing to Public and Private Repositories)
- JSON, XML, JPEG, TIFF, LAS / LAZ, SBET, ROS Bag Data Formats
- ROS (Robot Operating System)
- Parallel / Concurrent Computing

---

## Experience

### Innovation Engineer – Fugro

November 2022 – July 2025

- Researched, designed, built and iterated multi-sensor systems including cameras, LiDAR, INS, and other sensors and components.
- Used object-oriented design and programming (C# and C++) to develop critical data ingest, processing, calibration and geo-referencing software for multi-sensor systems.
- Took ownership of existing internal intrinsic and extrinsic camera calibration software and procedures.
- Developed new intrinsic and extrinsic calibration software and procedures for next generation cameras and multi-sensor systems.
- Assumed role of local calibration expert; Maintained calibration software codebases, updated documentation for software and procedures, and provided training and technical support to users.
- Researched and investigated innovative methods and software for intrinsic and extrinsic calibration of cameras, LiDAR, INS and other sensors.
- Performed troubleshooting and resolved bugs and deficiencies in production and pre-production software.
- Employed CI/CD pipelines in production codebases; Identified critical code and ensured extensive unit testing; Conducted code reviews of pull requests.
- Communicated complex information clearly, concisely and accurately, to co-workers, managers, and vendors. Properly documented work and wrote technical documentation for internal and external audiences.
- Enhanced technical and professional expertise and efficiency by using AI coding tools such as GitHub Copilot and Cursor, and studying modern programming techniques and design patterns.
- Translated user stories into technical designs and architectures and then implemented them; Ensured that after implementation the desired outcome was achieved.

---

### R&D Software Engineer – Mapsted

September 2016 – November 2022

- Designed key system components for location-based services such as mapping, positioning, navigation, marketing and analytics.
  - Used object-oriented design and programming (C++, C#, Java, Kotlin and Swift) to develop critical components for both server and client software.
  - Investigated innovative signal processing, adaptive filtering, optimization theory, machine learning, artificial intelligence, and probability theory and statistics techniques for mapping, positioning and navigation applications.
  - Found and resolved bugs and deficiencies in live production software. Performed maintenance, resolved issues and developed safeguards for production databases.
  - Communicated complex information clearly, concisely and accurately, to co-workers, managers, partners and clients. Properly documented work and wrote technical documentation for internal and external developer audiences.
  - Invented concepts and designs for new products/processes and submitted invention disclosures (lead inventor on five granted patents and contributor to several others).
  - Provided advanced technical support to clients and partners around the world, including on-site visits and demonstrations.
  - Managed and co-ordinated team members performing research, programming, and on-site operations tasks.
-

---

## Patents and Publications

- **US-9838850-B2** [Systems and Methods for Determining Indoor Location and Floor of a Mobile Device](#)
- **US-10070270-B1** [Mobile Device Localization Based on Spatial Derivative Magnetic Fingerprint](#)
- **US-10075935-B1** [Method and System for Positioning State Transition](#)
- **US-10123195-B1** [Method and System of Crowd-sourced Pedestrian Localization](#)
- **US-10206067-B1** [Method and System for Server Based Mobile Device Monitoring in Crowd-sourced Pedestrian Localization](#)
- **US-10257658-B1** [Self-learning Localization Data Repository](#)
- **US-10345452-B1** [Method and System for Mobile Device Localization Region in Trusted-GPS Region](#)
- **US-10349221-B1** [Method and System for Mobile Device Processing Timeout Based Localization](#)
- **US-10670728-B2** [Method and System for Crowd-sourced Trusted-GPS Region for Mobile Device Localization](#)
- **US-11356801-B1** [Method and System for Targeted Calibration](#)
- **US-11700504-B2** [Method and System for Calibration of an Indoor Space](#)
- **US-12225425-B2** [Method and System for Criterion-based Localization of Mobile Devices](#)
- **CA-3057025-A1** [Method and System for Crowd-sourced Map Feature Updating](#)
- **ISPRS Volume XLII-4** [Indoor Positioning Using WLAN Signal Fingerprint Matching and Path Evaluation with Retroactive Adjustment on Mobile Devices](#)
- **Masters Thesis** [Indoor Positioning Using WLAN Signal Fingerprint Matching and Path Evaluation with Retroactive Adjustment on Mobile Devices](#)

---

## Education

**M. Sc. Geomatics**      *York University*      Jan 2014 – Aug 2016

Research focused on indoor positioning and navigation on mobile devices, including the mapping and digitization of the indoor space. Developed a novel indoor positioning algorithm, implemented it on Android and deployed it across several buildings.

**B. Eng. Geomatics**      *York University*      Sep 2009 – Aug 2013

---